

PRASAD VICHARE

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WORK EXPERIENCE

University of Illinois Chicago

Dec 2024 - Present

Data Engineer | Graduate Assistant

chicago, USA

- **Improved financial data accuracy by 35%** by architecting a hybrid ingestion layer using **Azure Data Factory** to extract financial data from a **Microsoft SQL Server** backend (SoftMoney) into a **Microsoft Fabric** Lakehouse
- **Resolved complex orchestration dependencies** by implementing **Apache Airflow DAGs** to sequence **PySpark** heavy-lifting (JSON flattening/cleansing) before triggering **dbt** for final business logic modeling
- **Reduced ad-hoc reporting overhead by 40%** by building modular, version-controlled SQL models in **dbt**, allowing finance teams to access pre-computed Gold-layer datasets
- **Secured and optimized BI delivery** by implementing **Row-Level Security (RLS)** on SQL views and engineering Delta Lake datasets optimized for **Tableau** hyper-extracts, supporting university-wide variance analysis
- **Optimized pipeline cost and performance** by implementing **Change Data Capture (CDC)** on the source **SQL Server** to replace full-table refreshes with incremental loads into the Lakehouse

Reliance Jio Infocomm Pvt. Ltd

Sep 2020 - Jul 2024

Data Engineer | Deputy Manager

IND

- **Processed 15M-20M daily telecom events** by developing **Spark Structured Streaming** pipelines on **Azure Event Hub**, handling high-velocity telemetry for real-time spam detection
- **Guaranteed exactly-once processing** by implementing **idempotent writes** using Delta Lake operations and checkpointing to prevent duplicate records
- **Accelerated query performance on multi-terabyte datasets** through strategic **partition pruning** by event-date and applying **Z-ordering** on frequently joined subscriber columns
- **Standardized release reliability** by integrating **GitHub Actions** and **Azure DevOps** for **CI/CD**, automating syntax validation of **PySpark** scripts and **SQL** schema changes, which reduced manual review effort
- **Delivered KPI visibility** by transforming raw telemetry into curated datasets for **Power BI reporting**, enabling leadership to monitor network performance and make timely decisions

Quess Corporation

Jan 2019 - Sep 2020

Data Analyst

IND

- **Automated ~8 hours of weekly manual data prep** by engineering Python-based ingestion scripts to load raw operational records from **AWS S3** into **Snowflake**
- **Modernized operational reporting** by designing a **Snowflake** data warehouse using a Medallion architecture (Bronze, Silver, Gold), enabling scalable self-service analytics.
- **Optimized dashboard load times by 45%** through the implementation of **Snowflake clustering keys** and materialized views for high-concurrency reporting

SKILLS

- **Cloud & Platforms:** Microsoft Fabric, Azure Data Factory, Azure Event Hub, Databricks, Snowflake, AWS, Azure SQL Server, Azure DevOps
- **Data Engineering:** Apache Spark (PySpark), Kafka, Apache Airflow, dbt, Delta Lake, Structured Streaming, Change Data Capture (CDC), Medallion Architecture, Data Warehousing
- **Programming & Query:** Python (Pandas, NumPy), T-SQL, SparkSQL, PySpark, Shell Scripting., R, JAVA
- **Data Quality & Reliability:** Schema Evolution, Data Contracts, Partition Pruning, CI/CD (GitHub Actions, Azure DevOps), Z-Ordering
- **Analytics & Visualization:** Tableau (Hyper-extracts, RLS), Power BI (DAX, Semantic Modelling), Excel
- **Project Management:** Agile/Scrum, SDLC, Stakeholder Management, Jira, Microsoft Office Suite

PROJECT

Real-Time Fraud Detection System | Independent Research Project, UIC

- Built batch and real-time fraud detection pipelines using **ADF**, **Event Hubs (Kafka)**, and **Databricks** Structured Streaming, implementing medallion Lakehouse architecture with exactly-once processing.
- Developed stateful Spark streaming feature engineering (velocity windows, geospatial distance, terminal risk metrics), enabling low-latency fraud scoring and alert ingestion into Delta Lake.
- Implemented PSI-based **data drift monitoring** on prediction and feature distributions, triggering automated retraining signals and maintaining audit-ready Delta tables.

LLM-Based Multi-Agent Medical Report Analysis with Hallucination Control | AI Systems Project

- Architected a **7-stage analysis pipeline** using **LangChain** and **Google Gemini API**, orchestrating parallel execution of **3 specialist AI agents** via Python's **ThreadPoolExecutor** for concurrent inference
- Implemented a **multi-layer data validation framework**, including evidence grounding checks, cross-agent consistency validation, and confidence calibration scoring to quantify hallucination risk
- Built a **Flask-based backend generating structured JSON/PDF** outputs, enforcing schema compliance and traceable reasoning through **prompt engineering** and **temperature control**

EDUCATION

University of Illinois Chicago | **M.S., Management Information Systems**

Chicago, IL Aug 2024 - May 2026

University of Mumbai | **B.E., Electronics & Telecommunication**

Mumbai, IND Aug 2014 - May 2018

CERTIFICATIONS

- Databricks Certified Data Engineer Associate
- AWS Academy Data Engineering
- Microsoft Certified: Fabric Data Engineer Associate DP -700
- Google Data Analytics